A Preliminary Dataset and Analysis of Chatbot Use in a Critical Domain

A User Prompt Codebook

Blinded

April 9, 2026

Abstract

The rise of large language models has spurred demand for chatbots that provide efficient, centralized, and personalized knowledge access. We present a dataset of user interactions with a study advisor chatbot developed for prospective higher education students in Germany. The dataset includes anonymized student prompts from a prototype school deployment and a complementary "critical dataset" designed to test the chatbot's safeguards. By classifying and analyzing prompts, we identify patterns to guide database coverage, chat history use, and misuse prevention. This publicly available dataset, which will be periodically updated, supports research on chatbot safety, user behavior, and uncertainty detection by providing real-world German interactions. It enables the study of language-specific challenges and informs best practices for designing safe, user-centered chatbots.

Contents

1 Codebook Overview

This codebook for the dataset presented in *"A Preliminary Dataset and Analysis of Chatbot Use in a Critical Domain"* describes the two datasets, consisting of 441 (user prompts: `main_data.csv`) and 71 (additional critical prompts: `critical_data.csv`) user prompts. The data and this codebook will be periodically updated to add more data. We hope that the collected data can be helpful for researchers developing chatbots in critical domains such as education, and help them understand chatbot usage more.

2 Data Overview

Our main dataset consists of 441 individual prompts[1] by 70 individual users collected through three distinct piloting studies over the span of two months. We used slightly different configurations of the chatbot architecture, database, and hardware throughout, but all chatbots are based on the technology of open-source LLMs and RAG. As stated in the introduction, school classes used the chatbot during regular class hours. Participation was voluntary. To encourage free use of the chatbot and to avoid issues with data protection, no additional data about the students was gathered. The students visited grades 11/12, which corresponds to the ages 16-18, and are all on track to gain the qualification to enter higher education[2]. All classes were co-educational.

Our critical dataset consists of 71 critical prompts that were gathered by asking colleagues and university students to ask critical prompts. We used the dataset as a first evaluation (before the very first pilot) of how well our chatbot responds to these prompts. As already mentioned, prompts can be critical in several ways: they can be discriminatory, use profanities, entice the chatbot to answer in an unintended way (on purpose or not), or include genuine concerns that the chatbot should answer in a specific, sensitive way. The critical dataset, while being artificial in many ways, can also be used by others to evaluate their safeguards and is contrasted in this paper against the real critical prompts asked by students. Our datasets consist of ten variables, with each row being one prompt—prompts by the same student can be identified via an ID.

3 Features

In addition to the prompts, the dataset also includes four unique features. Aside from the three categorizations, we also include the SAE activations. For three features, we classify the prompts along three dimensions: the topic of the prompt, the type of the prompt, and whether the language used in the prompt indicates uncertainty. Each of these variables was coded by three independent coders; if there was disagreement, the category that two coders agreed on was used. Any cases where all coders disagreed were examined and finally assigned in a shared discussion.

[1] Note that we use the term *prompt* to mean user prompts throughout this paper.
[2] In German: Gymnasiale Oberstufe.

Variable	Description
PromptID	Overall PromptID, when combined with ID, sorted by which prompts were asked first by a user.
ID	User ID.
Prompt	User Prompt (German).
Prompt_EN	Translated user prompt (English).
german_word_count	Word count for German prompt.
english_word_count	Word count for English prompt.
activations_en_named	Top 10 SAE Features and their activation (based on English prompt).
topic	Topic of the prompt.
type	Prompt type, dialogue act.
certainty	Prompt certainty.

Table 1: Overview of dataset variables.

3.1 Topic Categories

We are interested in knowing the topic of the prompt to better assess whether our database is sufficient and to add information to the database first in those areas that most prompts are concerned with. In total, we identified ten types of topics: Next to critical prompts (which we have not further distinguished), we have prompts concerned with study programs, interests, jobs, grades, universities, the match of their interests with study programs, finances, general study-related questions, and other legitimate topics or utterances (e.g., "Hello").

Topic	Description
study_program	Questions about study programs and where they are offered.
critical	Critical questions such as rude comments or swear words.
interests	Users stating what they are interested in.
job	Questions about job possibilities.
grades	Questions about NC grades, entry exams, and qualifications.
uni	Questions about universities.
matching	Questions about how well someone matches with a study program or field.
general	General study orientation questions.
other	Other topics not covered by the above categories.
finance	Questions about financing studies, living expenses, etc.

Table 2: Overview of topic categories.

3.2 Type Categories

We further distinguish between six types of prompts based on the dialog act: question, answer (to the chatbot), request (telling the chatbot to do something), follow-up question (to a previous own question or chatbot answer), follow-up request, and statement. Here, we are particularly interested in whether students mostly use the chatbot in a conversational manner (indicated by many follow-up prompts) and whether they were mostly looking for specific information (indicated by questions) or rather converse freely about their interests and ideas.

Topic	**Description**
study_program	Questions about study programs and where they are offered.
critical	Critical questions such as rude comments or swear words.
interests	Users stating what they are interested in.
job	Questions about job possibilities.
grades	Questions about NC grades, entry exams, and qualifications.
uni	Questions about universities.
matching	Questions about how well someone matches with a study program or field.
general	General study orientation questions.
other	Other topics not covered by the above categories.
finance	Questions about financing studies, living expenses, etc.

Table 3: Overview of topic categories.

3.3 Certainty Categories

Finally, we distinguish between prompts that indicate that a user is certain about their topic and those indicating uncertainty. A (fictive) example for a certain prompt may be: *"I know I want to study Biology, and I'm really interested in the program at University ABC. Can you tell me more about how the program?"*, whereas a (fictive) example of an uncertain prompt may be: *"I guess I did well in my Biology class, so I could imagine doing something along those lines. I heard about the program at University ABC, but I'm unsure whether it's for me. Can you tell me more about the program?"*. In both cases, the chatbot should answer with details about the biology program at University ABC, but we want the chatbot to frame the answers differently. For example, the first prompt could be answered with basic information about the program and a question back about what the person is most interested in; the answer to the second prompt should perhaps include a link to an interest or study orientation test and ask the student about their general interests. We are most interested in observing whether uncertain prompts have certain topics or other aspects in common, to ensure (e.g., via chatbot-based persona prompting) that uncertain prompts are identified and handled with care.

Certainty	Description
certain	User prompt is certain or neutral.
uncertain	User seems unsure or uncertain and asks for help.

Table 4: Overview of certainty levels.

3.4 SAE

In addition to coding these variables, we also extracted the ten most important features of each prompt according to the sparse autoencoders (SAEs) trained by Neuronpedia[3]. An SAE is a type of neural network designed to learn efficient, interpretable representations of high-dimensional data. An SAE (like any autoencoder) consists of an encoder that maps inputs into a latent representation and a decoder that reconstructs the original input. In addition, SAEs impose a sparsity constraint on the latent layer, typically by penalizing the activation of neurons. This encourages the model to represent each input using only a small subset of latent features which then makes those features more interpretable. SAEs can be trained on internal activations of a larger model (here: a transformer model) with the goal of decomposing these dense representations into a set of sparse, more semantically meaningful features. Then, these features are mapped back into token space to generate interpretable features. To identify the most important features of a prompt, one feeds the prompt into the chosen SAE, and the most important features (according to the magnitude of the activations) are returned. By looking at the features that activate consistently for our prompt classes, we can interpret those features as encoding salient properties of these prompt classes. Utilizing Neuronpedia, we extracted the ten most important features from activations of the model llama3.3-70b-it using the residual stream source 50-resid-post-gf for both the English translations and the original German prompts. However, based on qualitative inspection, we limited further analysis to the features derived from the English translations, as these tended to be more interpretable and semantically meaningful.

In the dataset, the collected SAEs are represented as a list of the top 10 features and their activation for the English prompt.

4 Additional Figures

[3]https://www.neuronpedia.org/, last visited April 08, 2026

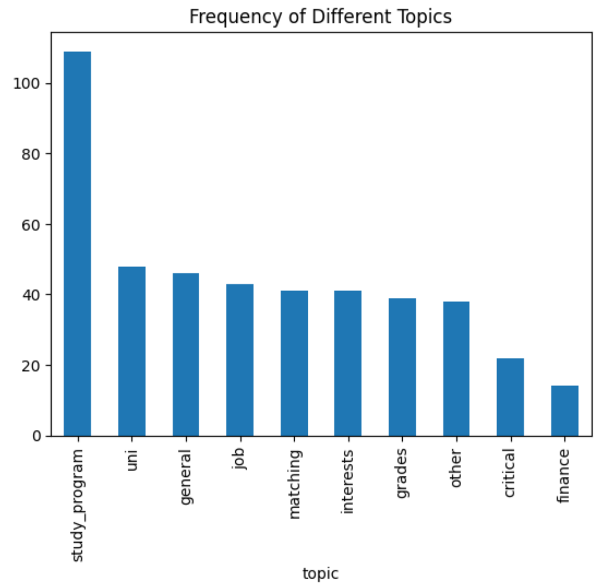

Figure 1: Frequency of different topics of prompts.

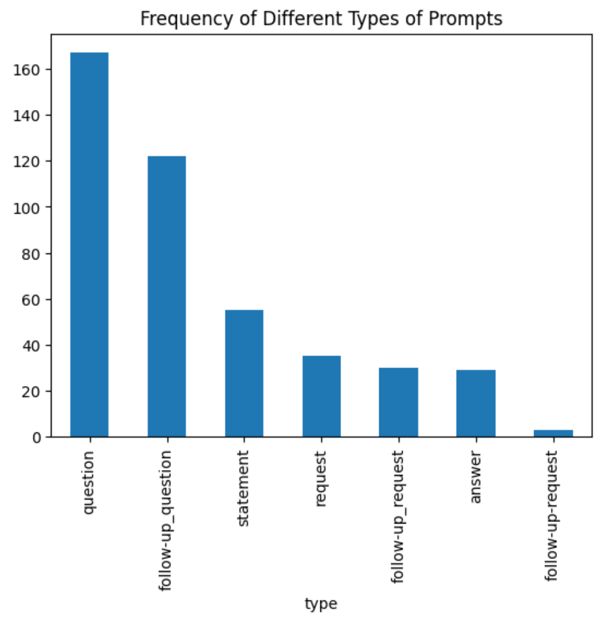

Figure 2: Frequency of different types of prompts.

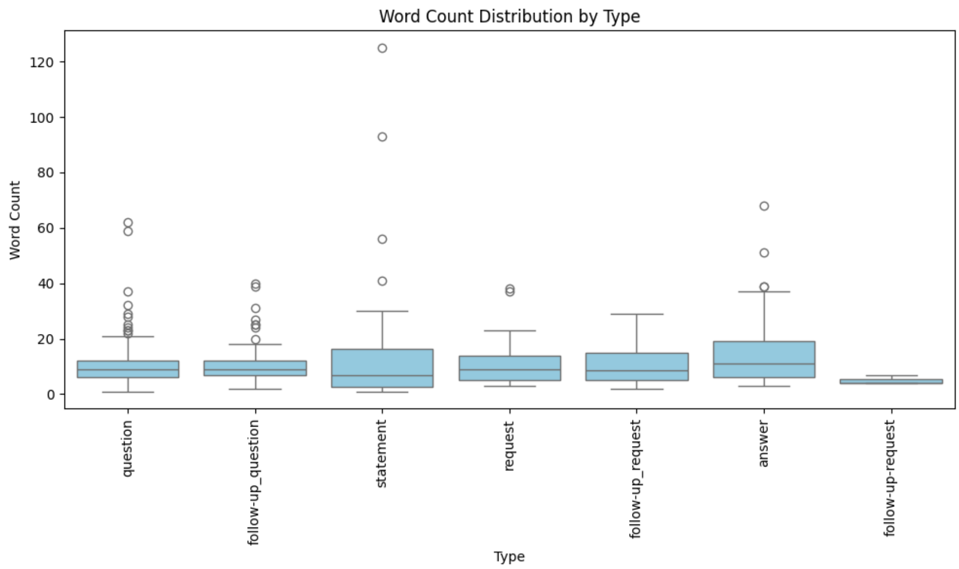

Figure 3: Word count distribution by type.

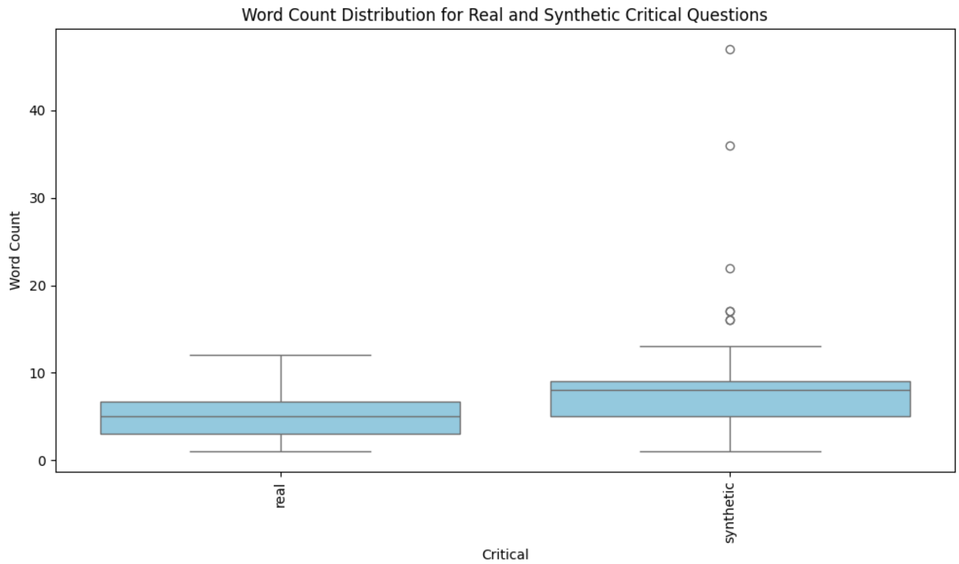

Figure 4: Critical prompt word count grouped by datasets.